

Investigating a vowel shift in progress in Santiago Laxopa Zapotec

Mykel Loren Brinkerhoff

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This article investigates a vowel shift in progress in Santiago Laxopa Zapotec's vowel system. Evidence from Pillai Scores and Bhattacharyya's Affinity show that the Sierra Norte Zapotec vowel /o/ is rising to /u/. Male speakers of Santiago Laxopa Zapotec have the largest divergence between [o] and [u] productions.

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1 Introduction

1.1 Vowel splits, mergers, and shifts

1.1.1 Previous reseach about splits, mergers, and shifts

1.1.2 Quantification through Pillai Scores and Bhattacharyya's Affinity

1.2 Santiago Laxopa Zapotec

Santiago Laxopa Zapotec (SLZ; endonym *Dille'xhunh Laxup*) is an Sierra Norte Zapotec variety and belongs to the Zoogochan Dialect group ([Smith-Stark 2007](#)). SLZ is spoken by approximately 1000 speakers in the pueblo of Santiago Laxopa ([Brinkerhoff 2025](#), [Brinkerhoff & McGuire 2025](#)). Most previous research has focused on the semantics and syntax of the language (see [Adler et al. 2018](#), [Sichel & Toosarvandani 2020, 2025](#), [Silva-Robles et al. 2022](#)).

Typologically, members of the Zoogochan Zapotec varieties typically have a four-vowel system consisting of the vowels / i, e, a, o / [see]

2 Methods

2.1 Data collection

Data was collected in the summer 2022 by myself and a fellow researcher in the pueblo of Santiago Laxopa, Ixtlán, Oaxaca, Mexico. Ten native speaker participants (five female; five male) performed a

word list elicitation task consisting of 72 items repeated three times in isolation and three times in the carrier sentence *Shnia' X chonhe lhas* [ʃn:ia' X tʃone ras] “I say X three times.” Recordings were made using a Zoom H4n handheld recorder (16-bit, 44.5 kHz).

2.2 Data processing

3 Analysis

4 Results

4.1 Vowel space

4.2 Vowel overlap

4.3 Vowel shift

5 Conclusions

6 Refernces

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